

Unsupervised Time-Aware Sampling Network With Deep Reinforcement Learning for EEG-Based Emotion Recognition

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Abstract—Recognizing human emotions from complex, multi-variate, and non-stationary electroencephalography (EEG) time series is essential in affective brain-computer interface. However, because continuous labeling of ever-changing emotional states is not feasible in practice, existing methods can only assign a fixed label to all EEG timepoints in a continuous emotion-evoking trial, which overlooks the highly dynamic emotional states and highly non-stationary EEG signals. To solve the problems of high reliance on fixed labels and ignorance of time-changing information, in this paper we propose a time-aware sampling network (*TAS-Net*) using deep reinforcement learning (DRL) for unsupervised emotion recognition, which is able to detect key emotion fragments and disregard irrelevant and misleading parts. Specifically, we formulate the process of mining key emotion fragments from EEG time series as a Markov decision process and train a time-aware agent

through DRL without label information. First, the time-aware agent takes deep features from a feature extractor as input and generates sample-wise importance scores reflecting the emotion-related information each sample contains. Then, based on the obtained sample-wise importance scores, our method preserves top- X continuous EEG fragments with relevant emotion and discards the rest. Finally, we treat these continuous fragments as key emotion fragments and feed them into a hypergraph decoding model for unsupervised clustering. Extensive experiments are conducted on three public datasets (SEED, DEAP, and MAHNOB-HCI) for emotion recognition using leave-one-subject-out cross-validation, and the results demonstrate the superiority of the proposed method against previous unsupervised emotion recognition methods. The proposed *TAS-Net* has great potential in achieving a more practical and accurate affective brain-computer interface in a dynamic and label-free circumstance.

Index Terms—Electroencephalography, affective brain-computer interface, emotion recognition, deep reinforcement learning, unsupervised learning.

I. INTRODUCTION

HUMAN emotion is a complex and dynamic process involving different levels of processing and integration [1], [2]. In emotion-related studies, how to accurately describe and further recognize human emotions has been a critical issue in the last decade. Recently, the developments in using electroencephalography (EEG) signals for emotion recognition have gained increasing attention from researchers with a various of backgrounds [3], [4], [5], [6], because EEG could provide a more direct and objective clue to understand and estimate emotional states. The current EEG-based emotion recognition studies are sample-based emotion recognition, which treats the EEG samples in a continuous data collection equally and assigns the fixed emotional label (trial-based groundtruth), without taking into account emotion dynamics. As a consequence, the highly viable emotional states and associated EEG signals are not explored comprehensively, which hinders our understanding of emotion dynamics and decreases the accuracy of emotion recognition. To solve these problems, a novel EEG-based emotion recognition approach is proposed in this paper to unsupervisedly and adaptively determine key emotion fragments from an EEG trial which would be then used for emotion recognition.

Currently, many existing EEG-based emotion recognition models have been built via supervised machine learning, which

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The source code is made available at <https://github.com/infinite-ao/TAS-Net>.

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can roughly fall into two categories. (1) Traditional machine learning-based methods. Based on the commonly used handcrafted EEG features, such as power spectral density (PSD) [7], differential entropy (DE) [8], differential asymmetry (DASM) [9], rational asymmetry (RASM) [10], and differential caudality (DCAU) [11], machine learning-based classifiers are built for emotion recognition [8], [12], [13]. For example, Alsolamy et al. [7] extracted the PSD features at different frequency bands and used a support vector machine (SVM) as a classifier to predict emotions during listening to Quran. To enhance emotion recognition performance, Atkinson et al. [14] introduced to use the minimum-Redundancy-Maximum-Relevance (mRMR) method to select the most emotion-related handcrafted features for modeling. However, due to the unstable and weak performance of the traditional machine-learning methods, more and more EEG-based emotion recognition models are developed by using (2) deep learning-based methods. EEG-based emotion recognition models with deep learning are able to automatically extract discriminant features and identify emotional states from EEG signals in an end-to-end manner [11], [15], [16], [17]. For example, Li et al. [18] proposed a bi-hemispheres domain adversarial neural network (BiDANN) based on the asymmetry between the left and right hemispheres of the brain, which made the data representation easy for emotion recognition by separately mapping the EEG data of the left and right hemispheres into discriminant feature spaces. Song et al. [5] proposed a dynamical graph convolutional neural network (DGCNN) with five types of handcrafted features as input, which dynamically learns the intrinsic relationships between different EEG channels represented by an adjacency matrix. Zhong et al. [19] proposed an EEG-based regularized graph neural network (RGNN) for performance enhancement. Li et al. [20] proposed to extract discriminative spatial-temporal EEG features via a region-to-global feature learning process. Tao et al. [21] developed an attention-based convolutional recurrent neural network (ACRNN) to adaptive estimate the importance of temporal and spatial information in EEG signals using the channel attention mechanism and self-attention mechanism. Compared with traditional machine learning-based methods, deep learning-based methods have achieved great success in EEG-based emotion recognition due to their powerful data representation ability.

Most of the existing deep learning methods for EEG-based emotion recognition require a large number of training samples with emotional labels. However, it is unrealistic to collect a large number of EEG signals from different participants and manually annotate each sample based on emotional labels. To increase the sample size, previous studies usually divided each EEG trial into samples with a fixed length of 1 s and annotated these samples with the fixed emotional label (trial-based groundtruth), which leads to the failure of neural networks to learn the true data distribution [11], [17], [22]. To alleviate the problem, Dura et al. [23] proposed an unsupervised emotional state recognition method that provided corrective labels through k-means clustering when supervised learning-based classifiers (i.e., SVM or CNN) made ambiguous or incorrect predictions. This approach is notable for its ability to improve recognition performance of supervised

classifiers by incorporating unsupervised clustering. In our previous work [24], [25], we proposed unsupervised learning-based methods to eliminate the model's reliance on labels. In particular, Liang et al. [25] developed a hybrid deep convolutional recurrent generative adversarial network named EEGFuseNet, which automatically characterized spatial and temporal dynamics from EEG signals in a self-learning paradigm and realized a possible way for unsupervised EEG feature learning in the affective brain-computer interface applications. However, the emotion dynamics in a trial were still underestimated. In a continuous emotion-evoking process, the elicited emotions are in a state of flux and the simultaneously collected EEG signals should not be assigned the fixed emotional label.

With the rapid development of deep learning technology, deep reinforcement learning (DRL) has been applied in many real-world scenarios, such as games [26], robotics [27], natural language processing [28], visual understanding [29], and neural architecture search [30]. Unlike the conventional machine learning method, DRL learns through the reward signals of actions. In other words, a reasonable reward function is essential for a specific task based on DRL. Mnih et al. [31] successfully approximated value function with a deep CNN, and enabled their agent to beat a human expert in several Atari games. In recent years, researchers have been exploring the use of DRL algorithms for emotion recognition tasks in various modal-specific scenarios, such as audio-visual [32], text [33], multimodal [34], face image [35], and EEG signal [36]. To develop an effective CNN architecture for EEG-based emotion recognition, Li et al. [37] utilized deep Q-learning in RL to determine the convolutional layer depth and the connectivity patterns between layers. In addition, as transformer has shown superior performance compared to CNN in various domains, Li et al. [36] also developed an automatic transformer neural architectures search (TNAS) framework for EEG-based emotion recognition. However, most of these methods ignored the emotion dynamics in a trial. On the other hand, Zhou et al. [38] proposed a novel DRL-based video summarization method, where a pre-trained CNN was employed to extract features of video frames, and an RL-based deep summarization network was designed to detect key video frames, which largely enhanced the model efficiency for video processing. Due to the ingenious design of its reward function, no labels were required, and only a one-step user interaction was necessary during the learning process. Subsequently, many keyframe detection works [39], [40], [41] are interpreted as a one-step Markov decision process (MDP) that utilized a straightforward and refined Policy Gradient algorithm for sequential decision-making tasks [42]. Inspired by [38], we introduce a novel time-aware sampling method to automatically detect key emotion fragments from the time-series EEG signals for effectively considering the emotion dynamics in a trial. In comparison to the work in [38], we make further improvements based on the characteristics of emotion occurrence, including agent design, reward improvement, and detection strategy. In other words, for each trial, only the EEG data located at the selected key emotion fragments are retained for further unsupervised emotion recognition, and other less informative data are discarded. Here, we take "EEGFuseNet +

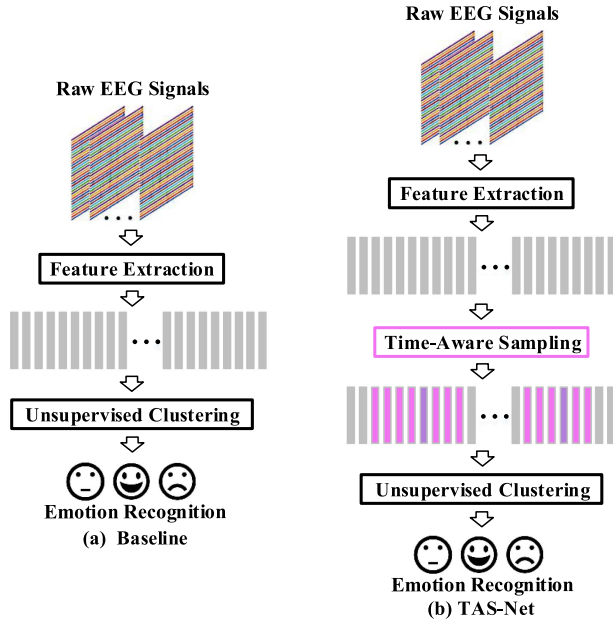


Fig. 1. Flowcharts of (a) the baseline method and (b) the proposed TAS-Net for unsupervised emotion recognition, respectively. In the proposed TAS-Net, the key emotion fragments (purple and light-purple parts) are first selected via a DRL-based time-aware sampling method, and then only the selected key emotion fragments are used for emotion recognition, as opposed to using all the samples in the baseline method.

Unsupervised Clustering” as the baseline method, and highlight the flowchart difference between the baseline method and the proposed TAS-Net in Fig. 1.

TAS-Net attempts to detect key emotion fragments in an unsupervised manner and further improves the model performance on EEG-based emotion recognition. First, an unsupervised deep feature extractor is employed to extract deep spatial and temporal features from raw EEG signals. Second, considering the non-stationary emotion in EEG time series, we propose a DRL-based time-aware sampling method to detect the most informative key emotion fragments from a trial. Here, a new time-aware agent that can better capture temporal dependencies is developed. It consists of a local time-based graph convolution network (GCN) and a global time-based bidirectional gated recurrent unit (BiGRU). Third, an unsupervised hypergraph decoding model is presented to categorize emotions based on the detected key emotion fragments. In summary, this work has three main contributions:

- We propose a novel unsupervised emotion recognition method based on deep reinforcement learning, which formulates the process of mining key emotion fragments from EEG time series as a one-step MDP. To the best of our knowledge, we are first to utilize deep reinforcement learning to locate informative emotion fragments for better unsupervised emotion recognition.
- We develop a novel time-aware agent, which consists of local time-based GCNs to learn the intrinsic relationship between adjacent segments and a global time-based BiGRU to capture long-term dependencies in EEG time series.

- We conduct comprehensive experiments on the emotion recognition task with the SEED, DEAP, and MAHNOB-HCI datasets. The proposed method not only achieves better performance than other unsupervised learning methods on the three public datasets, but is also comparable to some supervised learning methods under the same validation strategy. Additionally, compared to the baseline method, the performance using the proposed method is significantly improved.

II. METHODOLOGY

A. Overview

The framework of the proposed TAS-Net is shown in Fig. 2, in which three parts are included: (1) *deep feature extraction*: extract sample-based deep EEG features; (2) *time-aware sampling*: detect trial-based informative emotion fragments; and (3) *unsupervised clustering*: realize unsupervised emotion recognition using EEG signals. Here, the collected raw EEG signals in a trial are first segmented into a number of samples with a fixed length of 1 s, which are randomly sampled in mini-batches and fed into the deep feature extractor of the baseline method to extract sample-based deep EEG features. Second, the extracted sample-based deep EEG features are serialized into a trial-based representation and input to the time-aware agent (the first part in the time-aware sampling) to measure the contribution of each sample to the emotion dynamics in one trial. The output is a sequence of probabilities indicating how informative each sample is in referring to emotion dynamics in a trial. The EEG samples with high probabilities are then selected as the informative emotion samples in a trial. Considering that human emotion has “short-term continuity”, we further extract top- X continuous fragments centered on the selected informative emotion samples through the key fragment selection (the second part in the time-aware sampling). Finally, the extracted informative emotional fragments are used for emotion recognition using the unsupervised clustering method. More details about each module in the proposed TAS-Net are presented below.

B. Deep Feature Extraction

An unsupervised deep feature extraction network (EEGFuseNet) was proposed [25] to efficiently extract reliable features from high-dimensional raw EEG signals. Therefore, we employ the EEGFuseNet as our feature extractor to characterize sample-based deep EEG features $\{s_n\}_{n=1}^N$, N represents the number of mini-batch for each training. Specifically, EEGFuseNet was a hybrid deep encoder-decoder network architecture, which integrated a convolutional neural network (CNN), recurrent neural network (RNN), and generative adversarial network (GAN) in a smart and efficient manner. In the CNN-RNN based deep encoder-decoder network, the spatial and temporal dynamics in a time-series EEG data were efficiently characterized to represent the complex relationships of brain signals at different brain locations and at different time points. Based on the extracted spatial information by CNN, RNN was then adopted to measure the temporal information by exploring the

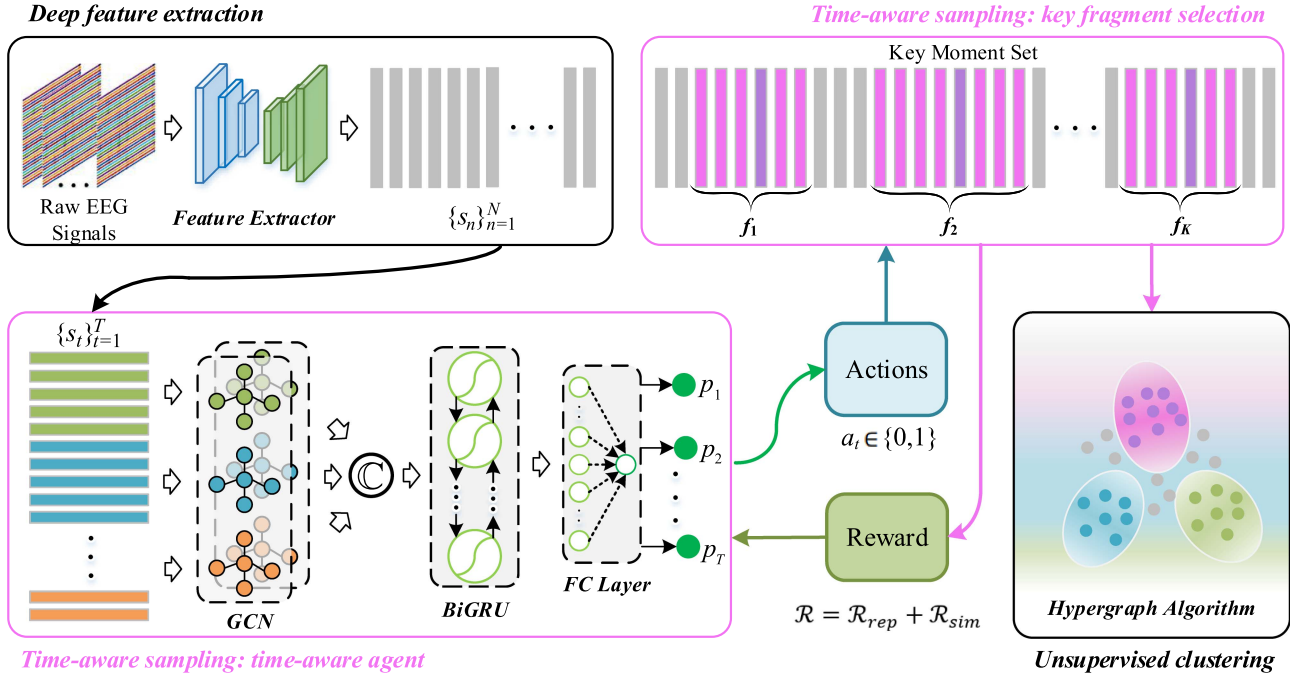


Fig. 2. Framework of the proposed TAS-Net, which includes *deep feature extraction*, *time-aware sampling*, and *unsupervised clustering*. $\{s_n\}_{n=1}^N$: sampled-based deep features, N represents the number of mini-batch. $\{s_t\}_{t=1}^T$: trial-based deep features, T denotes the time length of one trial. GCN: graph convolution network; $\odot$: concatenation; BiGRU: bidirectional gated recurrent unit; FC Layer: fully connected layer; p_t : action probability score; f_k : the detected key emotion fragments. Here, the detected key emotion fragments are composed of the detected key moments (dark purple) and the emotion offsets (light purple) calculated according to the nature of “short-term continuity” in human emotions. The final detected key emotion fragments are composed of both dark and light purple parts.

feature relationships at adjacent time points. To further benefit high-quality feature characterization, a discriminator was incorporated to enhance the generator (CNN-RNN based)’s performance. EEGFuseNet is a self-learning network without any reliance on labeled samples, which is suitable to solve unsupervised EEG processing.

C. Time-Aware Sampling

In a continuous emotion-evoking process, not all the time-point are equally important to the final elicited emotions. In other words, not all of the simultaneously collected EEG data in a trial should be treated equally using the fixed emotional label. Only a small percentage of EEG samples in a trial may contain significant and relevant emotion information, whereas the remaining data may represent less emotion information that could be discarded in emotion recognition. In this section, we present the proposed time-aware sampling method that adaptively highlights the most informative emotion fragments from each trial, without any need for label information.

Following the deep feature extraction, the time-aware agent in the proposed time-aware sampling approach interprets the key moment detection task as a one-step MDP. During the training phase, the time-aware agent interacts with an environment that provides rewards and takes actions a_t on whether a sample should be selected into the key moment set S by maximizing the total expected reward $\mathcal{R}$. In the inference stage, we retain the top- X continuous fragments centered on key emotion moments

for unsupervised clustering. Formally, we define the state, agent, action, and reward of the sequential decision-making process as follows:

State: The state is the key moment set S that is selected by the time-aware agent.

Agent: The time-aware agent is a sampler that takes a sequence of trial-based deep features $\{s_t\}_{t=1}^T$ as input and sequentially samples from a Bernoulli distribution to generate suitable actions for each moment. Specifically, to account for both “short-term continuity” and “long-term similarity” in human emotions, the time-aware agent is designed to consist of three components: a local time-based GCN, a global time-based BiGRU with a hidden size of 256, and a FC layer. Given the $\{s_t\}_{t=1}^T$ calculated by the deep feature extractor, the time-aware agent first divides the sequence of trial-based deep features into M successive segments ($m \in \{1, \dots, M\}$), which can be explained as follows:

$$C_m = \{s_i : i = (m-1) \cdot L + 1, \dots, \min\{T, m \cdot L\}\}, \quad (1)$$

where L denotes the time length of a successive segment, and T is the length of one trial. Then, the local time-based GCN takes these successive segments as input and estimates the local temporal features in parallel. After segmenting the local temporal features, we concatenate them along the time dimension and feed them into the global time-based BiGRU to generate the hidden state $\{h_t\}_{t=1}^T$. Finally, the FC layer ending with a sigmoid function predicts an action probability score p_t for each sample.

The action probability score is defined as

$$p_t = \sigma(W h_t), \quad (2)$$

where σ represents the sigmoid function, W is the trainable parameter of the FC layer, h_t is a concatenation of the forward hidden state h_t^f and the backward hidden state h_t^b .

Action: Action is defined as the sequence of moment-wise selection operation $A = \{a_t\}_{t=1}^T$, and is sampled based on the probability density of the Bernoulli distribution. Consequently, in this one-step MDP, actions do not exert a long-term impact. The action a_t is given as

$$a_t \sim \text{Bernoulli}(p_t), \quad (3)$$

where $a_t \in \{0, 1\}$ indicates whether t th sample in a trial should be selected in the key moment set or not. Therefore, the key moment set $\mathcal{S}$ can be defined as

$$\mathcal{S} = \{s_{y_i} \mid a_{y_i} = 1, i = 1, 2, \dots\}, \quad (4)$$

where y_i represents the indices of the selected samples.

Reward: To optimize the key moment selection of the time-aware agent and guarantee all the informative and discriminant points in a trial are covered, we introduce two reward signals to optimize the training process

$$\mathcal{R}(\mathcal{S}) = \mathcal{R}_{rep} + \mathcal{R}_{sim}, \quad (5)$$

where $\mathcal{R}_{rep}$ and $\mathcal{R}_{sim}$ denote the representativeness reward and the similarity reward, respectively. $\mathcal{R}_{rep}$ measures how well the selected key moment set $\mathcal{S}$ represents the original sequence of trial-based deep features, and it is defined as

$$\mathcal{R}_{rep} = \exp\left(-\frac{1}{T} \sum_{t=1}^T \min_{t' \in \mathcal{Y}} \|s_t - s_{t'}\|_2\right), \quad (6)$$

where $\mathcal{Y} = \{y_i \mid a_{y_i} = 1, i = 1, 2, \dots, |\mathcal{Y}|\}$. Through maximizing $\mathcal{R}_{rep}$, the time-aware agent is encouraged to select moments that are close to the cluster centers in the feature space. On the other hand, it is natural that the same emotion has similarity in different time periods, so we also introduce the similarity reward ($\mathcal{R}_{sim}$) to estimate the similarities among those selected key moment set $\mathcal{S}$, $\mathcal{R}_{sim}$ is given as

$$\mathcal{R}_{sim} = \frac{1}{|\mathcal{Y}|(|\mathcal{Y}| - 1)} \sum_{t \in \mathcal{Y}} \sum_{t' \in \mathcal{Y}, t' \neq t} \text{sim}(s_t, s_{t'}), \quad (7)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity of two vectors. With the similarity reward, the agent is encouraged to choose moments with high similarity in the feature space.

1) *Training With Policy Gradient:* Algorithm 1 shows the training details of the proposed time-aware agent. The goal of the time-aware agent is to learn a policy function $\pi_\theta(a|S)$ w.r.t parameters θ by maximizing the expected reward.

$$J(\theta) = \mathbb{E}_{p_\theta(a_{1:T})}[\mathcal{R}(\mathcal{S})], \quad (8)$$

where $p_\theta(a_{1:T})$ represents the probability distribution over possible action sequences, and $\mathcal{R}(\mathcal{S})$ is calculated by (5). According to the definition given above, the obtained action sequences could consist of different selection choices on the key emotion fragments. There are 2 types of action for each moment and the

Algorithm 1: Time-Aware Agent Training.

Input: Deep feature sequence $\{s_t\}_{t=1}^T$ from training set
Output: Parameters θ of the time-aware agent
1: **Initialize** θ with small random values
2: **for** $i \leftarrow 1, 2, \dots, \xi$ **do**
3: Input a deep feature sequence $\{s_t\}_{t=1}^T$ from training set
4: Compute action probabilities $\{p_t\}_{t=1}^T$
5: **for** $episode \leftarrow 1, 2, \dots, N$ **do**
6: Sample $\{a_t\}_{t=1}^T$ from policy π_θ
7: Compute reward $\mathcal{R}_n$
8: **end for**
9: Update θ with gradient $\nabla_\theta J(\theta)$ and $\nabla_\theta \mathcal{L}_{sampling}$
10: Update the moving average of rewards b for the corresponding deep feature sequence.
11: **end for**
12: **return** θ

exponential 2^T is computationally infeasible for the time-aware agent training. Thus, we employ the policy gradient method to efficiently compute the continuous action space.

Following the REINFORCE algorithm [38], [43], the derivative of the objective function $J(\theta)$ can be approximated as

$$\nabla_\theta J(\theta) \approx \frac{1}{E} \sum_{e=1}^E \sum_{t=1}^T (\mathcal{R}_e - b) \nabla_\theta \log \pi_\theta(a_t | h_t), \quad (9)$$

where E denotes the total number of training episodes, $\mathcal{R}_e$ is the reward computed at the e th episode. b is simply computed as the moving average of previous rewards to speed up the training. a_t and h_t represent the action and the hidden state at time t , respectively. Note that the policy function cannot update its parameters after each action. Instead, the update is only performed after the expected reward has been calculated. Therefore, we optimize the policy function via stochastic gradient-based method.

To avoid too many key moments selected for increasing the reward, we also introduce a regularization term $\mathcal{L}_{sampling}$ to limit the selection percentage for the key moment set. The regularization term is defined as

$$\mathcal{L}_{sampling} = \left\| \frac{1}{T} \sum_{t=1}^T p_t - \vartheta \right\|^2, \quad (10)$$

where ϑ is a scalar representing the percentage of key emotion fragments to be selected. Finally, we employ Adam optimizer to update the parameters θ of the policy function, which is calculated as

$$\theta = \theta - \varphi \nabla_\theta (-J + \beta \mathcal{L}_{sampling}), \quad (11)$$

where φ is learning rate, and β denotes a balance factor of the two terms.

2) *Key Fragment Selection in Inference:* Algorithm 2 shows the key fragment selection part in the proposed TAS-Net method. Given a deep feature sequence $\{s_t\}_{t=1}^T$ from a testing set, we feed it into the trained time-aware sampling model π to predict sample-level probabilities as importance scores. We first sort the

Algorithm 2: Time-Aware Agent Testing.

Input: Deep feature sequence $\{s_t\}_{t=1}^T$ from testing set
Output: Action probabilities $\{p_t\}_{t=1}^T$
1: **Initialize** θ from the trained time-aware agent
2: Input a deep feature sequence $\{s_t\}_{t=1}^T$
3: Compute importance scores $\{p_t\}_{t=1}^T$ through policy π_θ
4: Sort importance scores in descending order
5: Select the top- X continuous fragments according to (12)
6: Save the top- X continuous fragments for unsupervised clustering
7: **return** $\{p_t\}_{t=1}^T$

sample-wise importance scores in descending order and select top- X points as the key moments. Considering the nature of “short-term continuity” in human emotions, we further use the selected key moments to form top- X emotion fragments which are centered around the selected key moments. The emotion fragments are defined as

$$f_x = \{s_{\lceil \tilde{o}_x \rceil}, s_{\lceil \tilde{o}_x \rceil + 1}, \dots, s_{c_x}, \dots, s_{\lfloor \tilde{o}_x \rfloor}\}, x \in \{1, 2, \dots, X\}, \quad (12)$$

where s_{c_x} is the key moment of f_x , and $s_{c_x} \in \mathcal{S}$. $\lceil \cdot \rceil$ and $\lfloor \cdot \rfloor$ represent the ceiling and floor operators, respectively. $\tilde{o}_x^{\leftarrow}$ and $\tilde{o}_x^{\rightarrow}$ denote two real positive numbers, which are formulated as

$$\tilde{o}_x^{\leftarrow} = c_x - \tilde{o}_x \cdot o_{\max}, \quad (13)$$

$$\tilde{o}_x^{\rightarrow} = c_x + \tilde{o}_x \cdot o_{\max}. \quad (14)$$

where $\tilde{o}_x$ represents relative offset coefficient, which is equal to the importance score p_{c_x} at c_x th key moment. $o_{\max} \in \mathbb{Z} \geq 0$ is the maximum absolute offset to the left or right. Finally, we obtain top- X key emotion fragments for unsupervised emotion recognition.

D. Unsupervised Clustering Model

For a fair comparison with the baseline method, we also employ the hypergraph structure [44] to perform unsupervised emotion recognition. Hypergraph is widely regarded as an effective method to describe complex hidden data structures, so it has natural advantages in decoding EEG signals and completing emotion classification. In a hypergraph, one hyperedge could connect more than two nodes and reveal more complex hidden structures than a pairwise connection in a simple graph (one edge can only connect two nodes). In the implementation, we formulate a hypergraph as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, w)$, where $\mathcal{V}$ refers to vertices indicating the trial-based EEG representation obtained in the proposed time-aware sampling method and $\mathcal{E}$ is the constructed hyperedge information based on the $\mathcal{V}$. The unsupervised emotion recognition is realized by calculating the hypergraph Laplacian of the constructed hypergraph ($\mathcal{G}$) and solving it in an optimal eigenspace.

III. EXPERIMENTAL RESULTS

We conduct a comprehensive experiment to evaluate the effectiveness of the proposed TAS-Net in dealing with unsupervised

emotion recognition using EEG signals. In this section, we would first describe the datasets used in our experiments, followed by an introduction to experiment settings and evaluation metrics. Then, we would report the experimental results. The promising results demonstrate the effectiveness of the proposed method.

A. Datasets

We perform experiments on three publicly available EEG datasets, including SEED [11], DEAP [45], and MAHNOB-HCI [46]. All these datasets have been widely used as benchmarks for evaluating EEG-based emotion recognition algorithms.

- **SEED:** The SEED dataset contains EEG signals of 15 subjects, which were recorded at a 62-channel setting according to the international 10–20 system [47] when each subject watched 15 film clips. These film clips can evoke specific target emotions (positive, neutral, or negative), and each film clip is about 4 minutes long.
- **DEAP:** The DEAP dataset contains EEG signals of 32 subjects, which were recorded at a 32-channel setting according to the international 10–20 system when each subject watched 40 music videos. These music videos were all 60 seconds in duration, and each was given corresponding subjective feedback on different emotional dimensions (valence, arousal, dominance, and liking). The participants were asked to rate different emotion dimensions on a scale of 1 to 9 after watching each music video. To cross-compare with the baseline method and other methods, we use a fixed threshold of 5 for each emotion dimension to discretize the subjective feedback into two classes (low and high).
- **MAHNOB-HCI:** The MAHNOB-HCI dataset contains EEG signals of 30 subjects, which were recorded at a 32-channel setting according to the international 10–20 system when each subject watched 20 video clips. These film clips were selected to evoke emotions and the subjective feedback was given using a score in the range of 1 to 9. In the model evaluation, a fixed threshold of 5 is used to discretize the subjective feedback into binaries for each emotion dimension (valence, arousal, dominance, and predictability).

B. Experiment Settings and Evaluation Metrics

The proposed TAS-Net is implemented in the PyTorch platform. Here, we set the L in (1) to 16, the ϑ in (12) to 0.5, the β in (13) to 0.01, and the number of episodes N to 5. The dimension of the edge feature in the GCN is 32, and the dimension of the hidden state in the GRU cell is 256. The above hyper-parameters are set according to the relevant literature [48]. The proposed TAS-Net uses an Adam optimizer with an initial learning rate of 10^{-4} and a weight decay of 10^{-5} . Besides, the training is stopped when it reaches a maximum number of epochs (100 in our case).

Inconsistent model validation strategies have been used in previous EEG-based emotion recognition methods, which can

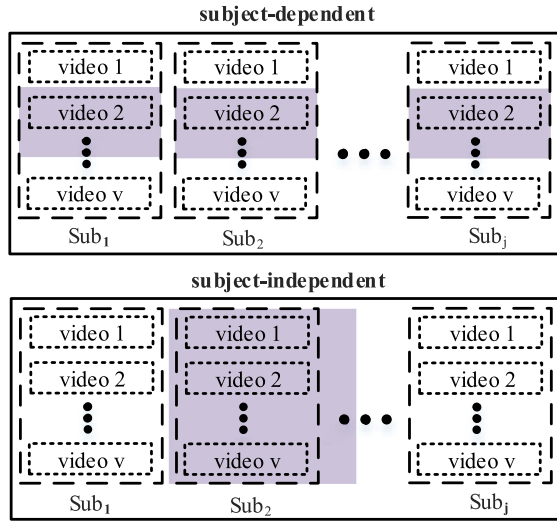


Fig. 3. Two validation strategies for EEG-based emotion recognition. The subject-dependent validation strategy divides data sets (purple represents the test set and the rest is the training set) within the subject, which can be subdivided into two categories: video-level K-fold cross-validation (CV) and video-level leave-one-out cross-validation (LOOCV). The subject-independent validation strategy divides data sets (purple represents the test set and the rest is the training set) among subjects, which can be subdivided into two categories: subject-level K-fold CV and subject-level LOOCV. v refers to the number of videos used in the experiment, and Sub_j represents j th subject.

be divided into two categories, subject-dependent and subject-independent. Fig. 3 shows the details of data partitioning for the above two validation strategies. The subject-level LOOCV strategy avoids information leakage, so it has lower recognition performance compared to other validation strategies. For a fair comparison with the baseline method, we mainly use the subject-level LOOCV strategy. In each validation round, only one subject's all data are used as the testing data, and the rest of the other subjects' data are treated as training data.

In our experiments, we employ two performance metrics (accuracy P_{acc} and F1-Score P_f) to quantitatively evaluate the obtained emotion recognition results. P_{acc} is a common evaluation metric for the classification task, and P_f is more able to measure the classification performance of datasets with imbalanced classes. Note that all the results reported below are an average performance across all subjects in different datasets.

C. Evaluation on SEED Dataset

We compare the proposed TAS-Net with the state-of-the-art emotion recognition methods. Table I reports the comparison results, in which the adopted learning methods and cross-validation protocols are clearly stated. In general, compared to the unsupervised learning-based methods, the supervised learning-based methods would achieve better emotion recognition performance, as both data and label information are involved in the model training process. In the table, the best results of each sub-experiment are highlighted in bold. The TAS-Net performance (P_{acc}) on the two-class (positive and negative) classification and three-class (positive, neutral, and negative) classification are 82.34% and 63.10%, respectively. Compared

TABLE I
EMOTION RECOGNITION PERFORMANCE (%) ON SEED DATASET

Methods	Classification Task	P_{acc}	P_f
Supervised		Subject-Dependent	Video-level LOOCV
DBN [11]	Three-Class	86.08	-
GSCCA [22]	Three-Class	82.96	-
BiDANN [18]	Three-Class	92.38	-
DGCNN [5]	Three-Class	90.40	-
R2G-STNN [20]	Three-Class	93.38	-
RGNN [19]	Three-Class	94.24	-
Supervised with Transfer Learning		Subject-Independent	Subject-level LOOCV
TCA [15]	Three-Class	63.64	-
BiDANN [18]	Three-Class	83.28	-
DGCNN [5]	Three-Class	79.95	-
R2G-STNN [20]	Three-Class	84.16	-
RGNN [19]	Three-Class	85.30	-
JDA [17]	Three-Class	88.28	-
Supervised without Transfer Learning		Subject-Independent	Subject-level LOOCV
JDA [17] (source domain only)	Three-Class	58.23	-
Unsupervised		Subject-Independent	Subject-level LOOCV
Baseline [25]	Two-Class	80.83	82.03
	Two-Class	59.06	-
	Two-Class	82.34	82.32
TAS-Net (Ours)	Three-Class	63.10	-

The best results of each sub-experiment are highlighted in bold.

to the baseline method, the proposed method enhances the average accuracy of the above two classification tasks by 1.87% and 6.84%, respectively. The performance difference between the two methods is further evaluated by the paired t-test, and the obtained p -values are reported in Fig. 4(a), which shows that the proposed method has significant differences ($p < 0.05$) compared to the baseline method. On the other hand, our method and the supervised transfer learning method (TCA [15]) achieve similar performance. In general, the results shown in Table I and Fig. 4(a) prove that the proposed method can perform more accurate unsupervised emotion recognition on the SEED dataset.

D. Evaluation on DEAP Dataset

Similar to the existing studies, a binary classification task is conducted to evaluate the emotion recognition performance of each emotion dimension on the DEAP dataset. Table II lists the corresponding results in terms of emotion recognition accuracy P_{acc} and F1-Score (P_f). The results demonstrate the superior performance of our method over other unsupervised learning approaches. Furthermore, our proposed method exhibits comparable effectiveness when compared to certain supervised methods (i.e., Li et al. [49], Koelstra et al. [45] and Torres-Valencia et al. [53]), where the recognition accuracies (P_{acc}) of valence, arousal, dominance, and predictability are 57.84%, 60.51%, 63.17% and 67.32% and the corresponding F1-Score (P_f) values are 71.80%, 72.64%, 75.32%, and 79.74%. Compared with the baseline method, our method improved the average accuracy of the four emotion dimensions by 2.48%, 3.35%, 2.37%, and 2.17%, with significant difference ($p < 0.05$) as shown in Fig. 4(b). As a result, our method can be regarded as significantly better than the baseline method on the DEAP dataset. Also, compared with the other existing unsupervised EEG decoding method [24], our method also achieves significant performance gains on all emotion dimensions.

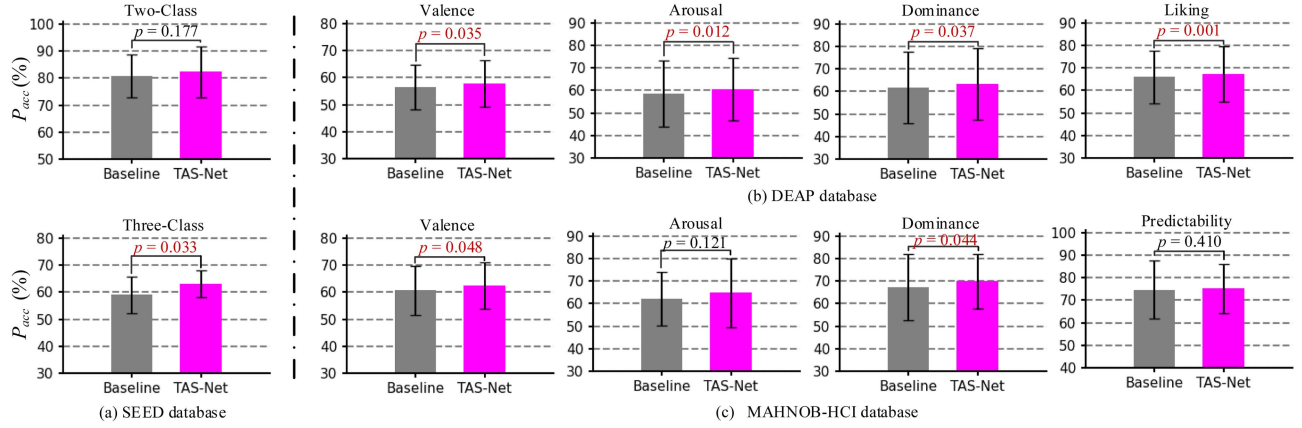


Fig. 4. Obtained p -values of the paired t-test between the proposed TAS-Net and the baseline method in terms of P_{acc} . The error bars represent the standard deviation of the obtained experimental results.

TABLE II
EMOTION RECOGNITION PERFORMANCE (%) ON DEAP DATASET

Methods	Valence P_{acc}	P_f	Arousal P_{acc}	P_f	Dominance P_{acc}	P_f	Liking P_{acc}	P_f
Supervised	Subject-Dependent							
Li et al. [49]	58.40	-	64.20	-	65.80	-	66.90	-
Chen et al. [50]	67.89	67.83	69.09	68.96	-	-	-	-
DGCNN [5]	93.50	-	92.55	-	93.50	-	-	-
NAS [37]	97.94	-	97.74	-	97.82	-	-	-
TNAS [36]	98.66	-	98.68	-	98.67	-	-	-
Supervised	Subject-Dependent							
Video-level LOOCV								
Koelstra et al. [45]	57.60	56.30	62.00	58.30	-	-	55.40	50.20
DT-CWPT [51]	64.30	-	66.20	-	68.90	-	70.20	-
EMD [52]	69.10	-	71.99	-	-	-	-	-
Supervised	Subject-Independent							
K-fold CV								
Torres-Valencia et al. [53]	58.75	-	55.00	-	-	-	-	-
Atkinson and Campos [14]	73.14	-	73.06	-	-	-	-	-
Liu et al. [54]	69.90	-	71.20	-	-	-	-	-
Supervised	Subject-Independent							
Subject-level LOOCV								
Shahnaz et al. [55]	64.71	74.94	66.51	76.68	66.88	76.67	70.52	81.94
DGCNN [5]	59.29	-	61.10	-	-	-	-	-
ATDD-LSTM [56]	69.06	-	72.97	-	-	-	-	-
Supervised & Unsupervised	Subject-Dependent							
K-fold CV								
SVM [23]	80.88	-	82.30	-	-	-	-	-
CNN [23]	75.27	-	86.85	-	-	-	-	-
Unsupervised	Subject-Dependent							
Video-level LOOCV								
Liang et al. [24]	56.25	61.25	62.34	60.44	64.22	64.80	66.09	77.52
Unsupervised	Subject-Independent							
Subject-level LOOCV								
Liang et al. [24]	54.30	53.45	55.55	52.77	57.03	52.84	58.91	59.99
Baseline [25]	56.44	70.83	58.55	72.00	61.71	74.32	65.89	78.46
TAS-Net (Ours)	57.84	71.80	60.51	72.64	63.17	75.32	67.32	79.74

The best results of each sub-experiment are highlighted in bold.

E. Evaluation on MAHNOB-HCI Dataset

Similarly, a binary classification task is conducted on the MAHNOB-HCI dataset to evaluate the emotion recognition performance of each emotion dimension. As shown in Table III, our proposed method achieves a comparable performance compared to the other supervised methods, where the recognition accuracies (P_{acc}) of valence, arousal, dominance, and predictability are 62.51%, 64.69%, 69.84% and 75.11% and the corresponding F1-Score (P_f) values are 74.64%, 60.60%, 79.34%, and 84.10%. Compared with the baseline approach, our method improved the average accuracy of the above four affective dimensions by 3.08%, 4.24%, 4.11%, and 0.64%, respectively. The corresponding statistical difference analysis is reported in

TABLE III
EMOTION RECOGNITION PERFORMANCE (%) ON MAHNOB-HCI DATASET

Methods	Valence P_{acc}	P_f	Arousal P_{acc}	P_f	Dominance P_{acc}	P_f	Predictability P_{acc}	P_f
Supervised	Subject-Dependent							
Video-level LOOCV								
Zhu et al. [57]	55.72	51.44	60.23	57.77	-	-	-	-
Supervised	Subject-Independent							
Subject-level LOOCV								
Soleymani et al. [46]	57.00	56.00	52.40	42.00	-	-	-	-
Huang et al. [58]	62.13	-	61.80	-	-	-	-	-
LRFS [59]	69.93	71.22	67.43	68.58	-	-	-	-
Unsupervised	Subject-Independent							
Subject-level LOOCV								
Baseline [25]	60.64	72.18	62.06	62.05	67.08	76.65	74.63	83.61
TAS-Net(Ours)	62.51	74.64	64.69	60.60	69.84	79.34	75.11	84.10

The best results of each sub-experiment are highlighted in bold.

Fig. 4(c). It shows, compared with the baseline method, our method achieves significant performance improvement on two emotional dimensions (valence and dominance), while achieving similar performance on the other two emotional dimensions (arousal and predictability).

IV. DISCUSSIONS

To fully investigate the effectiveness and efficiency of the proposed TAS-Net, we conduct multiple sets of ablation experiments, including comparing different components, evaluating the defined rewards, evaluating the existing clustering methods with or without TAS-Net, and analyzing the parameter effect. Further, to quantitatively evaluate the key emotion fragment detection performance using TAS-Net, we introduce temporal intersection over union (tIoU) [48] to calculate the recall rate of samplings. In the performance comparison with different clustering methods, we add normalized mutual information (NMI) as another performance metric to check the corresponding clustering quality. Due to space limitations, all experimental analyses in this section are based on the SEED dataset.

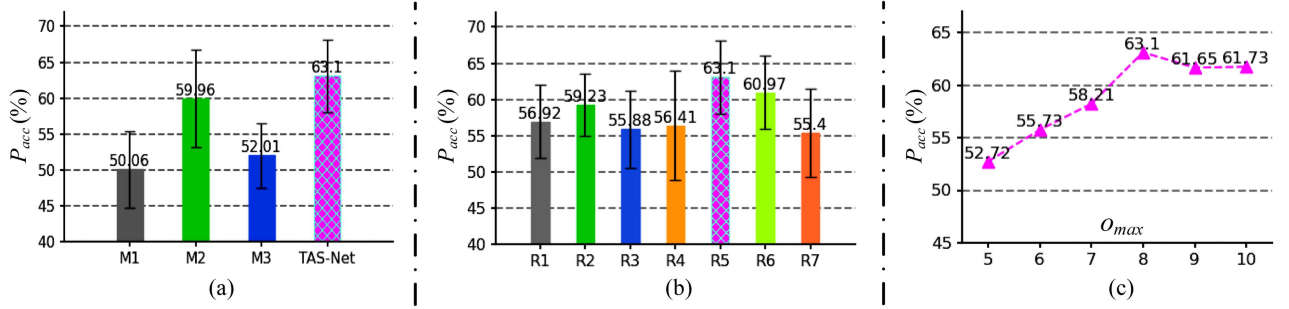


Fig. 5. Emotion recognition performance (%) under different methods and parameters comparison. (a) Comparison results of different components. M1, M2, and M3 represent that the agent is defined as a FC layer, a local time-based GCN topped with a FC layer, and a global time-based BiGRU topped with a FC layer, respectively. (b) Comparison results of different reward functions. R1, R2, R3, R4, R5, R6 and R7 indicate the reward function given as $\mathcal{R}_{rep}$, $\mathcal{R}_{sim}$, $\mathcal{R}_{div}$, $\mathcal{R}_{rep} + \mathcal{R}_{div}$, $\mathcal{R}_{rep} + \mathcal{R}_{sim}$, $\mathcal{R}_{rep} + 1/\mathcal{R}_{ED}$ and $\mathcal{R}_{rep} + \mathcal{R}_{ED}$, respectively. (c) Comparison results of different values of parameter o_{max} . The error bars represent the standard deviation of the obtained experimental results.

A. Ablation Study

We conduct ablation experiments to evaluate the contribution of each component in the proposed TAS-Net. The compared methods are: (i) M1: the agent is a FC layer, which produces the action probability. (ii) M2: the agent is a local time-based GCN topped with a FC layer; (iii) M3: the agent is a global time-based BiGRU topped with a FC layer [38]. Fig. 5(a) shows the obtained accuracies when different methods are used for unsupervised emotion recognition. The best performance is achieved by seamlessly combining local time-based GCN and global time-based BiGRU (the proposed TAS-Net). Compared with M1, M2 and M3, our method improves the average accuracy P_{acc} by 26.05%, 5.24%, and 21.32%, respectively. Therefore, we believe that it is effective to model the time-aware agent by considering that human emotion has not only “short-term continuity” but also “long-term similarity”.

B. Effectiveness of Rewards

To explore the optimization effect of reward functions in the proposed TAS-Net, we set different reward functions for comparative experiments. Here, $\mathcal{R}_{rep}$ is to measure how well these selected key emotion fragments can represent the whole trial’s emotional information. $\mathcal{R}_{div}$ is another commonly used reward function that measures the diversity of the generated summary by evaluating the difference between the selected key frames in the feature space [38]. However, since signals evoked by the same emotion should be similar, we believe that replacing $\mathcal{R}_{div}$ with $\mathcal{R}_{sim}$ would better align with this theory. $\mathcal{R}_{sim}$, in this case, would encourage the agent to choose moments that are similar to each other in the feature space, promoting a more coherent representation of the emotional state. Fig. 5(b) shows the obtained unsupervised emotion recognition accuracies using different reward functions. R1, R2, R3, R4 and R5 indicate the reward function given as $\mathcal{R}_{rep}$, $\mathcal{R}_{sim}$, $\mathcal{R}_{div}$, $\mathcal{R}_{rep} + \mathcal{R}_{div}$ and $\mathcal{R}_{rep} + \mathcal{R}_{sim}$, respectively. It is found that the reward function $\mathcal{R}_{sim}$ achieves better performance than $\mathcal{R}_{div}$, which indicates that the similarity reward function is more suitable for the key emotion fragment detection task. Compared with the reward function $\mathcal{R}_{rep}$, when the reward function is composed of $\mathcal{R}_{rep}$

and $\mathcal{R}_{div}$, the recognition performance decreases, while when the reward function is composed of $\mathcal{R}_{rep}$ and $\mathcal{R}_{sim}$, the recognition performance increases, which proves that the same emotion has similarity in different time periods. To further evaluate our reward design, we also utilize the euclidean distance to measure the pairwise feature distribution distance in the key moment set. When the sum of the distances $\mathcal{R}_{ED}$ of pairwise feature distributions is smaller, the similarity is stronger, and vice versa. Therefore, we also compare two reward functions (i.e., R6 and R7), which represent $\mathcal{R}_{rep} + 1/\mathcal{R}_{ED}$ and $\mathcal{R}_{rep} + \mathcal{R}_{ED}$ respectively. In Fig. 5(b), we can find that the emotion recognition result obtained by using R6 is better than that of R7, which further confirms the validity of our motivation for designing the reward function.

C. Performance Comparison Using Different Clustering Methods

The purpose of this study is to design a sampling model that can efficiently localize short-term continuous fragments with correlated emotions in EEG time series, so as to achieve better unsupervised emotion recognition performance. Theoretically, the sampling set can effectively improve the emotion recognition performance of different clustering methods. We employ six decoding models for this comparative experiment, including a simple graph based method, principal component analysis (PCA) and K-means clustering method (PCA+Kmeans), K-nearest neighbors (KNN) algorithm, robust continuous clustering method (RCC), directed graph based agglomerative algorithm (AGDL), and hypergraph algorithm. The simple graph completes unsupervised clustering by measuring pair-wise relationships. K-means is one of the outstanding representatives of unsupervised clustering, and KNN is one of the simplest classification algorithms in supervised clustering. RCC is a fast, simple, and efficient high-dimensional clustering algorithm that achieves decoding of highly mixed clusters by optimizing a clear global objective. AGDL is a graph-based agglomerative algorithm, which quantifies the stability of clustering results by computing the product of average indegree and average outdegree. Table IV reports the unsupervised recognition accuracies

TABLE IV
EMOTION RECOGNITION PERFORMANCE (%) WITH THE STATE-OF-THE-ART CLUSTERING METHODS USING SUBJECT-INDEPENDENT LOOCV STRATEGY ON SEED DATASET, UNDER THE CONDITIONS WITHOUT (w/o) AND WITH (w) THE PROPOSED TIME-AWARE SAMPLING METHOD

Methods	Sampling	Two-Class			Three-Class	
		P_{acc}	P_f	NMI	P_{acc}	NMI
Simple graph	w/o	64.16	64.15	0.1728	35.82	0.0018
	w	65.96	66.22	0.1896	56.16	0.2800
PCA+k-means	w/o	56.12	65.27	0.0658	40.09	0.0686
	w	60.64	61.23	0.0879	58.22	0.2868
KNN	w/o	60.87	39.18	0.1139	42.04	0.1132
	w	79.72	80.04	0.4237	52.99	0.2504
RCC [60]	w/o	55.52	66.24	0.0601	38.27	0.0752
	w	58.95	69.38	0.0869	42.56	0.1276
AGDL [61]	w/o	51.09	67.63	0	34.47	0
	w	53.45	67.43	0.0481	35.89	0.0399
Hypergraph	w/o	80.83	82.03	0.4381	59.06	0.3569
	w	82.34	82.32	0.4135	63.10	0.3750

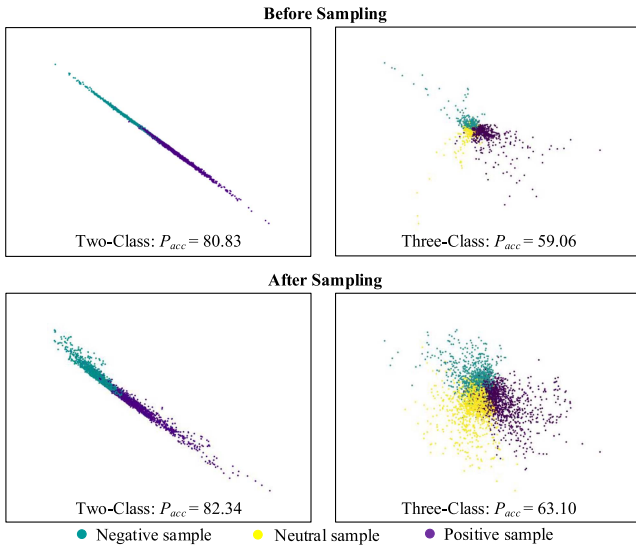


Fig. 6. Before and after sampling, the t-SNE visualization results of the hypergraph clustering algorithm on the two-class and three-class emotion recognition tasks.

of different clustering methods without or with the time-aware sampling method. Upon using the time-aware sampling method, all clustering methods demonstrate significant performance improvements in the three-class unsupervised emotion recognition task. Additionally, the majority of the two-class classification results are also enhanced. This shows the effectiveness of the proposed time sampling method in unsupervised emotion recognition using EEG signals. The t-SNE visualization results presented in Fig. 6 show that the hypergraph clustering algorithm performs better at distinguishing samples after applying the time-aware sampler.

D. Parameter Analysis

We further quantitatively evaluate the effectiveness of the proposed TAS-Net under different parameters. There are two important hyper-parameters in our method: the number K of

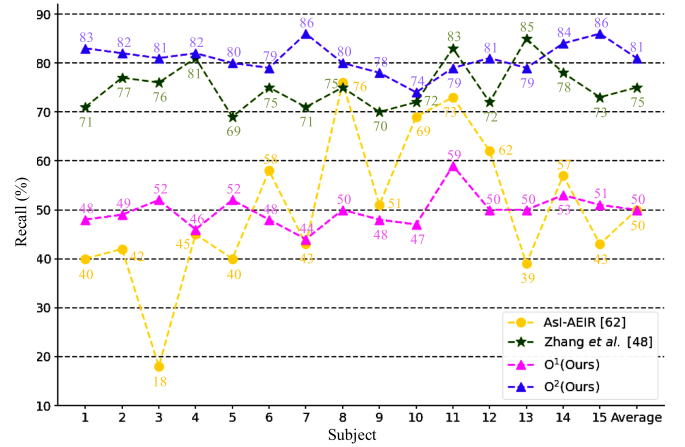


Fig. 7. Average recall (%) obtained by different methods on 13 video stimuli. O^1 and O^2 represent our method based on subject-independent LOOCV and subject-dependent LOOCV, respectively.

selected fragments and parameter o_{max} representing the maximum absolute offset to the left or right. The setting of K can affect the recall rate of sampling and the accuracy of emotion recognition, while o_{max} determines the maximum length of a selected fragment. Following [48], we set K to 10 and set o_{max} to 5, 6, 7, 8, 9, 10. As shown in Fig. 5(c), our method achieves the best performance when o_{max} is 8, which means that the max length of a fragment with coherent stronger emotion is 16. According to the experimental results, we clearly find that our method is sensitive to this parameter when o_{max} is less than 8, while the results obtained by our method are relatively stable when o_{max} is greater than 8.

E. Performance on Emotion Localization

Recently, Zhang et al. [48] proposed a supervised learning based hierarchical self-attention network on SEED dataset, which incorporates emotion localization task for improving emotion recognition performance. Similarly, we also extend our proposed TAS-Net on emotion localization application under an unsupervised learning manner. In the emotion localization task, EEG signals corresponding to each video stimulus were segmented, and then twenty people were asked to score each fragment according to the corresponding emotion to obtain the ground truth labels. In addition, the recall is used as the evaluation metric for this task, and the threshold of tIoU (the overlap rate in the timeline between the ground truth fragments and the top- X selected fragments f_x) is set to 0.5. Note here that, for cross-comparison with [48], both subject-independent LOOCV and subject-dependent LOOCV are used for emotion localization in our method. The results presented in Table V indicate that the emotion localization performance of our proposed method using subject-dependent LOOCV is significantly higher compared to that obtained using subject-independent LOOCV. We believe that individual differences are a difficulty in cross-subject research, and the emotion localization task is no exception. Additionally, Fig. 7 illustrates the average recall of emotion localization achieved by various methods on

TABLE V
THE RECALL (%) OF EMOTION LOCALIZATION TASK ON SEED DATASET

Subject ID	#1		#2		#3		#4		#5		#6		#7		#8		#9		#10		#11		#12		#13		Average	
	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²	O ¹	O ²
Subject1	33	100	50	75	0	80	67	67	100	100	67	83	60	80	50	100	50	75	17	83	25	75	50	75	60	80	48	83
Subject2	33	100	75	100	60	80	50	67	100	100	17	67	40	60	25	100	75	100	17	83	75	75	50	75	20	60	49	82
Subject3	67	100	25	75	80	80	50	67	100	100	67	83	40	100	25	75	75	75	33	67	25	75	25	75	60	80	52	81
Subject4	0	67	50	100	60	80	33	67	67	100	17	100	40	80	75	75	50	75	50	67	75	100	25	75	60	80	46	82
Subject5	100	67	75	75	20	80	33	83	100	100	67	83	60	80	50	100	25	75	0	67	50	75	50	75	40	80	52	80
Subject6	0	100	75	75	60	80	33	67	67	100	50	67	60	80	75	75	25	75	33	83	25	75	75	75	40	80	48	79
Subject7	33	100	50	75	40	80	33	83	33	100	33	83	20	80	0	75	75	100	50	83	75	75	75	100	60	80	44	86
Subject8	67	100	50	75	40	80	33	67	100	100	33	67	40	80	75	100	25	75	33	67	75	75	25	75	60	80	50	80
Subject9	67	67	50	75	80	80	33	67	100	100	50	83	60	60	50	75	0	75	33	83	25	75	50	100	20	80	48	78
Subject10	33	100	50	75	60	80	33	67	67	100	17	50	40	40	50	75	50	75	33	67	25	75	75	75	80	80	47	74
Subject11	100	67	25	100	60	80	33	67	67	100	50	67	60	80	25	75	75	75	50	83	50	100	75	75	100	60	59	79
Subject12	67	100	50	75	80	80	67	67	67	100	50	83	80	60	0	75	25	100	50	83	75	75	25	75	20	80	50	81
Subject13	33	67	75	75	80	80	17	67	67	100	67	83	40	80	100	75	50	75	33	67	25	100	25	75	40	80	50	79
Subject14	100	100	25	75	20	80	33	83	67	100	67	83	60	80	50	100	75	100	50	67	50	75	50	75	40	80	53	84
Subject15	67	100	100	75	20	100	33	83	100	100	17	67	60	80	50	75	50	100	33	83	25	75	50	100	60	80	51	86
All Subjects	53	89	55	80	51	81	39	71	80	100	45	77	51	75	47	83	48	83	34	76	47	80	48	80	51	77	50	81

O¹ and O² represent our method based on subject-independent LOOCV and subject-dependent LOOCV, respectively

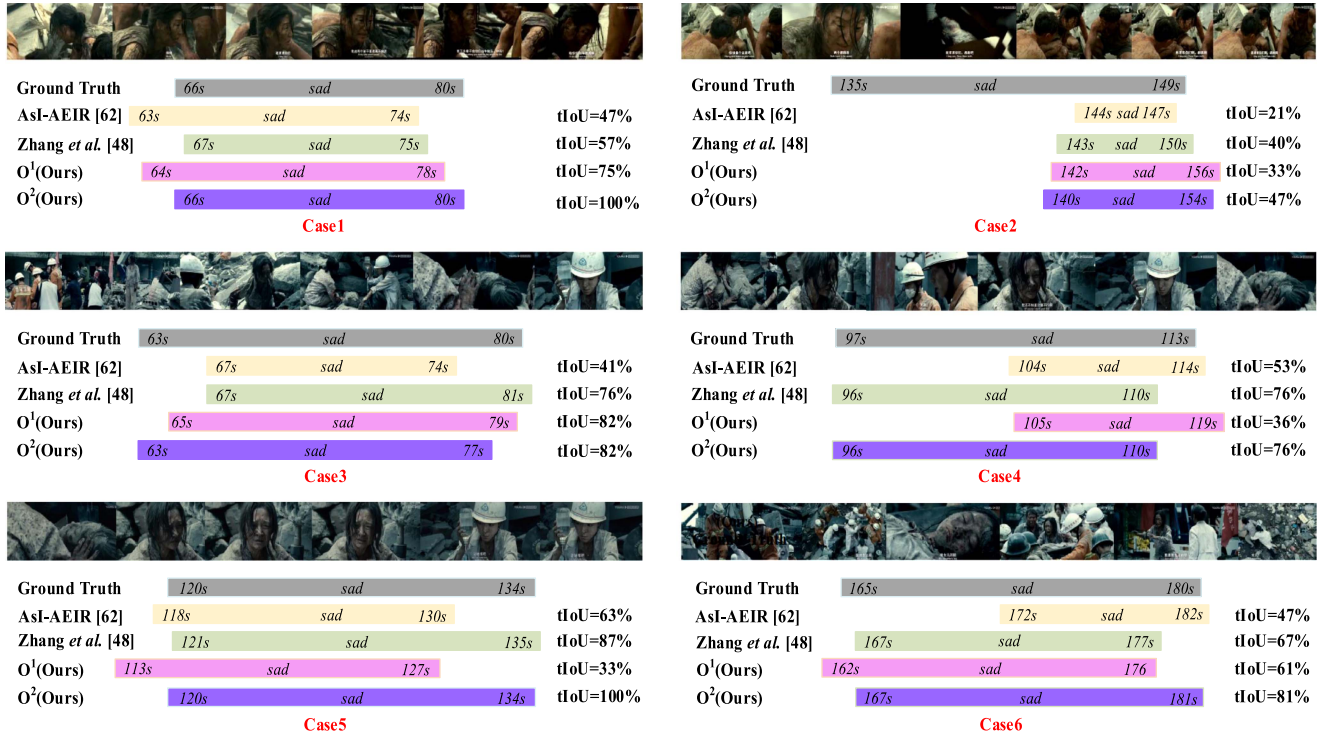


Fig. 8. Visualization of selected key fragments in different videos using different methods. O¹ and O² represent our method based on subject-independent LOOCV and subject-dependent LOOCV, respectively.

the 13 video stimuli. Compared with the other two emotion localization methods that employ subject-dependent LOOCV, our method achieves the highest performance. When subject-independent LOOCV is used (which is more difficult than subject-dependent LOOCV due to the individual differences), our model's subject-independent LOOCV results are comparable with AsI-AEIR [62]'s subject-dependent LOOCV results. In other words, our model is more effective and efficient in emotion localization under both subject-dependent LOOCV and subject-independent LOOCV. For a more intuitive comparison, we also visualize the emotion localization results when different methods are used. As shown in Fig. 8, the ground truth labels with the localization results and tIoU values are reported, which

shows the possibility to detect key emotion fragments even when label information is missing.

F. Limitations and Future Outlook

Building upon the observation that emotions evoked during a continuous process are not uniformly distributed in time, this study proposes the DRL-based TAS-Net to adaptively identify the salient moments of strong emotions and comprehensively evaluates its efficacy. Despite its promising results, this work also faces some limitations, such as computational time, hyper-parameter selection, and suboptimal outcomes attributed to separate training. First, the proposed method can be further

optimized to enable real-time high-efficient computing. The current TAS-Net architecture consists of a GCN and a BiGRU, which have 0.2369 M and 0.6917 M parameters, respectively. During training, the proposed method requires approximately 0.665 GB of GPU memory, and each training and real-time evaluation process takes about 35~46 minutes. Testing an EEG time series, on the other hand, takes only about 0.4~0.6 seconds. While these results indicate that the training and testing speed of the proposed TAS-Net is acceptable, the algorithm should be further optimized to reduce its complexity for more widespread applications. Second, in order to obtain more accurate strong emotion locations and boost unsupervised emotion recognition performance, the method inevitably introduces several hyperparameters. We set the values of these hyperparameters based on empirical observations from previous studies. However, in future research, we plan to employ a multi-objective optimization algorithm [63] to select the optimal hyperparameters. Third, the proposed method trains the feature extractor and the time-aware sampler separately, which may limit the model's ability to learn end-to-end optimized parameters. In the future, an end-to-end model will be developed to further improve unsupervised emotion recognition performance.

V. CONCLUSION

This paper develops an unsupervised time-aware sampling network (TAS-Net) based on deep reinforcement learning to adaptively perceive the location of strong emotions in continuously collected EEG signals, thereby improving the performance of unsupervised emotion recognition. To avoid a flat importance score for each sample-level feature in the EEG time series, the proposed TAS-Net accomplishes more efficient key emotion fragment detection by seamlessly combining local time-based GCN and a global time-based BiGRU. The TAS-Net, for the first time, applies deep reinforcement learning to detect key emotion fragments in EEG signals for more accurate unsupervised emotion recognition. Experiments on three challenging public datasets show that our TAS-Net achieves excellent performance. The performance in detecting key emotion fragments from the continuously collected EEG signals has been verified, and the superiority of emotion recognition has been demonstrated. Since the label information is not required by the proposed new method, TAS-Net is a fully unsupervised method that could be easily extended to other EEG-related applications and be utilized to automatically extract essential information from time-series EEG data.

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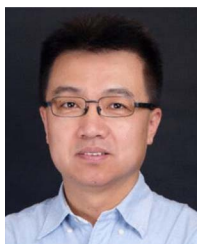
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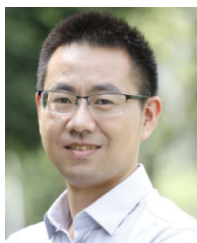
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